

Introduction and Background

A student's performance in Mathematics significantly dictates their trajectory in STEM-related careers and their future competitiveness in the global labor market. Beyond individual success, Mathematics proficiency is a key metric for institutional prestige. As a woman in STEM, I am personally committed to bridging the gender gap in technical subjects. This led me to conduct a pro-bono consultancy project for my alma mater—a C2-level girls' high school in Machakos County—where Mathematics has historically been the lowest-performing subject.

Despite years of teacher-led interventions and board-level brainstorming, student performance remained stagnant. As a research professional and data analyst, I applied a data-driven approach to diagnose why previous interventions failed. Initial trend analysis revealed a consistent pattern of decline starting immediately after the first end-of-term assessment in Form 1. This decline affected all students, including those who entered the school with high KCPE scores, suggesting that the issue was systemic rather than a lack of foundational ability.

Methods

I designed an exploratory study to evaluate student perspectives across two primary domains: **Student-Related Factors** (Interest, Attitude, and Study Habits) and **Teacher-Related Factors** (Teaching Style, Expectations & Support, and Communication & Availability). Each domain was assessed via a 15-item survey using a 5-point Likert scale to measure levels of agreement.

Using a random sampling technique to ensure representation, I selected 56 students from a Form 3 cohort of 250 across five streams. To maintain the integrity of the data, the survey was administered anonymously, and students were assured of confidentiality to encourage honest reporting.

Data processing was conducted using a dual-tool strategy: **Microsoft Excel** was used for initial cleaning and descriptive visualization, while **R** was utilized for inferential analysis. I also modernized the delivery of these insights into a **Power BI dashboard** to allow for real-time filtering by stream and specific performance metrics.

Findings

The descriptive data showed consistent student-related trends across all five streams; however, teacher-related factors showed significant variance. Among student-related barriers, **attitude** emerged as the primary hindrance to performance across the board. Conversely, "Teacher Support and Expectations" showed fluctuation between different streams, suggesting that the classroom environment is not uniform across the cohort. However, inferential analysis did not indicate statistical significance in these variations.

Initial results showed positive correlations between study habits and performance, and student attitudes and performance. However, the regression analysis yielded a counterintuitive finding. The resulting model — $\text{Performance} = 42.25 - 7.53(\text{Attitude}) + 7.63(\text{Study Habits})$ — revealed that students who reported a "positive attitude" actually tended to score lower. This suggests a **confidence-competence gap**, where overconfidence may be preventing students from identifying and correcting their academic weaknesses.

The model's R^2 value of **0.202** indicates that these factors explain approximately 20% of the variance in performance. While statistically significant, this leaves 80% of performance drivers unexplored, highlighting the need for deeper investigation into socio-economic and environmental factors. Notably, analyses of teacher-related factors did not yield statistically significant correlations with performance, further centering student attitude as the primary area for intervention.

Conclusions and Recommendations

To improve Mathematics outcomes, the school must shift its focus toward targeted student-related interventions.

1. **Early Intervention:** Mindset and attitude-shaping programs must be implemented in Form 1 to arrest the decline in performance before it becomes a multi-year trend.
2. **Addressing the Confidence Gap:** Current interventions should be re-evaluated to ensure they are building actual technical competence rather than just a false sense of security.
3. **Data-focused Approach:** The implementation of a data-based tracking system will allow stakeholders to monitor trends in performance and habits, allowing for proactive, rather than reactive intervention.
4. **CBE Alignment:** As a STEM-pathway school under the Competency-Based Education (CBE) framework, evidence-based decision-making is critical, and these findings provide a baseline for learner-centered interventions that address psychological barriers faced by girls in technical subjects, which will further ensure the effectiveness of the school's STEM environment.